Cyclic Scale Factors and the Limits of Temporal Periodicity in FLRW Cosmology

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Abstract

This note gives a mathematically consistent reformulation of the cyclic-universe idea suggested by a closed FLRW cosmology. We prove a precise reconstruction statement: any smooth periodic scale factor that stays strictly positive can be made into an exact solution of the Friedmann equations by an appropriate effective density and pressure. We also prove the converse point that periodicity of the scale factor does not, by itself, imply periodic time. The strongest defensible conclusion is therefore cyclic expansion and contraction, not cyclic time as a derived result.

1 Introduction

The starting point is the standard FLRW ansatz for a homogeneous and isotropic universe,

$$ds^2 = -c^2dt^2 + a(t)^2\left(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right),$$

where $a(t)$ is the scale factor and $k \in \{-1, 0, +1\}$ is the spatial curvature parameter. The Friedmann equations are

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \tag{1}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}, \tag{2}$$

with continuity equation

$$\dot{\rho} + 3H\left(\rho + \frac{p}{c^2}\right) = 0. \tag{3}$$

The main correction to the informal cyclic-time argument is conceptual: a periodic $a(t)$ is a statement about the geometry of the solution, while periodic time would require an additional global identification $t \sim t + T$. The field equations do not impose that identification.

2 A rigorous cyclic ansatz

A mathematically safe cyclic model should keep the scale factor positive. A convenient choice is

$$a(t) = a_{\min} + \frac{a_{\max} - a_{\min}}{2}\left[1 - \cos(\omega t)\right], \qquad \omega = \frac{2\pi}{T}, \tag{4}$$

with $0 < a_{\min} < a_{\max}$. Then $a(t)$ is smooth, periodic, and bounded below away from zero.

Proposition 1 (Direct reconstruction)**.** *Fix any C^2 function $a(t) > 0$. For any chosen k and Λ, define*

$$\rho(t) = \frac{3}{8\pi G}\left[\left(\frac{\dot{a}}{a}\right)^2 + \frac{kc^2}{a^2} - \frac{\Lambda c^2}{3}\right]. \tag{5}$$

Then $\rho(t)$ satisfies the first Friedmann equation identically. If one also defines

$$p(t) = -\frac{c^2}{4\pi G}\frac{\ddot{a}}{a} + \frac{\Lambda c^4}{12\pi G} - \frac{c^2}{3}\rho(t), \tag{6}$$

then the second Friedmann equation is also satisfied identically.

Proof. The first identity is obtained by solving the first Friedmann equation for $\rho(t)$. Substituting this expression into the second Friedmann equation and solving for $p(t)$ yields the stated formula. Hence the pair (ρ, p) reconstructs the chosen scale factor exactly. □

Corollary 1 (Existence of cyclic FLRW solutions)**.** *Every smooth periodic scale factor with $a(t) > 0$ can be realized as an exact FLRW solution for a suitable effective fluid.*

This corollary is the strongest statement one can make without specifying a microphysical matter model. It proves that cyclic expansion and contraction is mathematically consistent at the level of the homogeneous isotropic background.

3 Why periodic scale factor does not mean periodic time

Theorem 1 (No inference from periodic $a(t)$ to cyclic time)**.** *Assume $a(t)$ is periodic with period T. Then, by itself, this does not imply that time is periodic or that spacetime contains the identification $t \sim t + T$.*

Proof. The function $a(t)$ is a scalar field on a time coordinate line. Its periodicity means only that $a(t + T) = a(t)$ for all t. The time parameter still ranges over $\mathbb{R}$ unless one independently quotients the manifold by a discrete translation. No Friedmann equation, by itself, performs that quotient. Therefore periodic geometry of the solution is not equivalent to periodic topology of time. □

Remark 1. *One may still postulate a globally cyclic time manifold, but that is an extra axiom. It is not derived from the cosmological field equations used here.*

4 Worked example

Take

$$a(t) = a_0(1 + \varepsilon\cos(\omega t)), \qquad 0 < \varepsilon < 1. \tag{7}$$

Then $a(t) \geq a_0(1 - \varepsilon) > 0$. The Hubble parameter is

$$H(t) = \frac{\dot{a}}{a} = -\frac{\varepsilon\omega\sin(\omega t)}{1 + \varepsilon\cos(\omega t)}. \tag{8}$$

The reconstructed density becomes

$$\rho(t) = \frac{3}{8\pi G}\left[\frac{\varepsilon^2\omega^2\sin^2(\omega t)}{(1 + \varepsilon\cos(\omega t))^2} + \frac{kc^2}{a_0^2(1 + \varepsilon\cos(\omega t))^2} - \frac{\Lambda c^2}{3}\right]. \tag{9}$$

This exhibits explicitly how a periodic scale factor can be supported by an effective fluid, provided the resulting $\rho(t)$ and $p(t)$ remain physically acceptable for the chosen model.

5 What can be claimed honestly

The rigorous claim is not that “time closes on itself” because $a(t)$ is periodic. The rigorous claim is:

(1) a closed FLRW universe can be modeled with a periodic, strictly positive scale factor;

(2) the corresponding matter sector can be reconstructed exactly from the Friedmann equations;

(3) periodicity of $a(t)$ does not imply periodic time without an extra global identification.

This is already a nontrivial mathematical result, and it is the correct form in which the cyclic idea should be presented.

6 Conclusion

The original intuitive picture becomes mathematically consistent once the equations are written in standard FLRW form and the scale factor is chosen to remain positive. What is proved is cyclic cosmological evolution of the scale factor. What is not proved, and cannot be proved from the Friedmann equations alone, is that time itself is periodic.

References

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[3] V. Mukhanov, *Physical Foundations of Cosmology*, Cambridge University Press (2005).